

Domain and range



1

a

- i Domain = 8, 1, 3, 9, 2
- ii Range = 5, 3, 4, 1, 9
- iii Function

c

- i Domain = 4, 8, 2, 7, 3
- ii Range = 4, 8, 9, 1
- iii Function

e

- i Domain = 7, 8, 5, 3
- ii Range = 1, 9, 3, 2, 6
- iii Not

g

- i Domain = $3 \leq x \leq 7$
- ii Range = $5 \leq y \leq 8$
- iii Function

i

- i Domain = 5
- ii Range = $2 \leq y \leq 6$
- iii Not

k

- i Domain = $-7 \leq x \leq 7$
- ii Range = $0 \leq y \leq 7$
- iii Function

m

- i All real x
- ii $y \geq 0$
- iii Function

b

- i Domain = 2, 3, 6, 8
- ii Range = 3, 8, 1, 7, 2
- iii Not

d

- i Domain = 1, 6, 3, 8, 2
- ii Range = 3, 2, 7, 1
- iii Function

f

- i Domain = 2, 4, 6, 8, 10
- ii Range = 2, 4, 6, 8, 9
- iii Function

h

- i Domain = $2 \leq x \leq 7$
- ii Range = 5
- iii Function

j

- i Domain = $2 \leq x \leq 9$
- ii Range = $3 \leq y \leq 8$
- iii Not

l

- i Domain = $-9 \leq x \leq 9$
- ii Range = $-9 \leq y \leq 9$
- iii Not

n

- i $x \geq 0$
- ii All real y
- iii Not

2

a

- i Domain = $(-\infty, \infty)$
- ii Range = $(-\infty, 2]$

c

- i Domain = $[0, 6]$
- ii Range = $[0, 7]$

b

- i Domain = $(-\infty, \infty)$
- ii Range = $[-2, \infty)$



3

a False

b False

c True

d True

4

a $(-\infty, \infty)$

b $(-\infty, \infty)$

c $[-6, 6]$

5

a 1

b No

c All real x

d $y > 0$

e $y = 128$

f $x = 8$

6

a 16

b $y \leq 16$

c All real x

7

a 0

b $y \geq 0$

c $x > 7$

8

a

i Minimum

ii Minimum value = -1

iii $y \geq -1$

b

i Maximum

ii Maximum value = -3

iii $y \leq -3$

9

a $x \geq -1$

b No

10

a Concave up

b 5

c $y \geq 5$

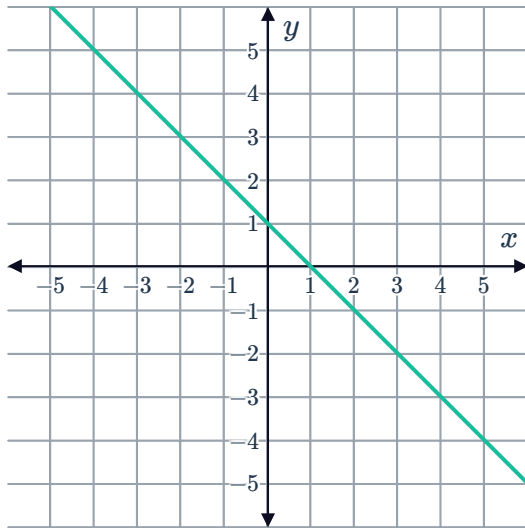


Graphing functions

11

a

i

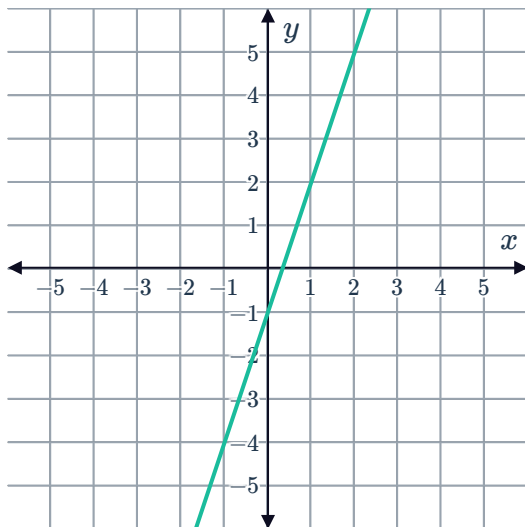


ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

b

i

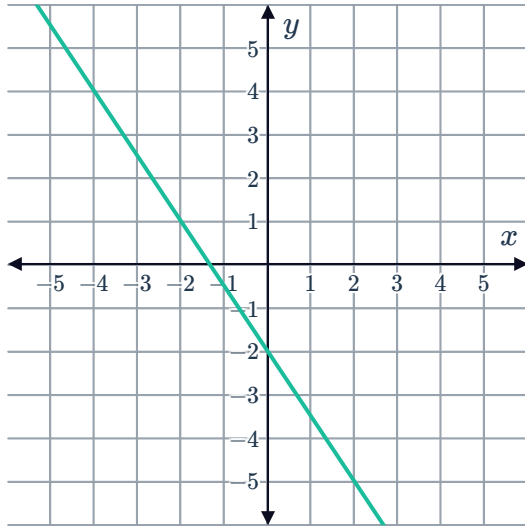


ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

c

i



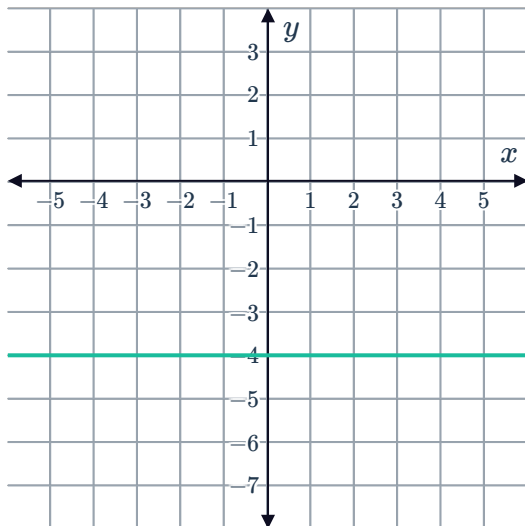
ii

$(-\infty, \infty)$

iii $(-\infty, \infty)$

d

i



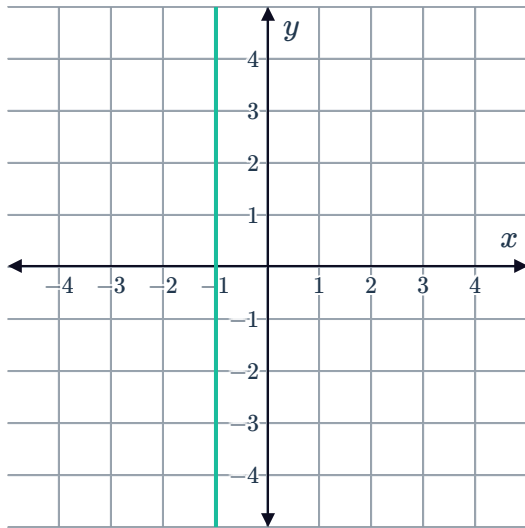
ii

$(-\infty, \infty)$

iii -4

e

i



ii

-1

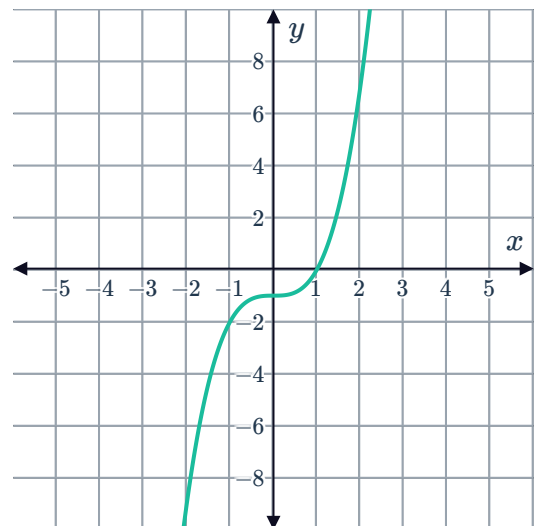
iii $(-\infty, \infty)$

12

a

x	-2	-1	0	1	2
y	-9	-2	-1	0	7

b



c Yes

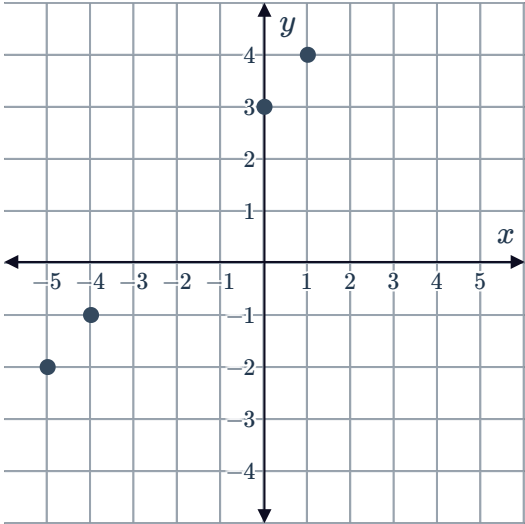
13

a

x	-5	-4	0	1
$f(x)$	-2	-1	3	4



b

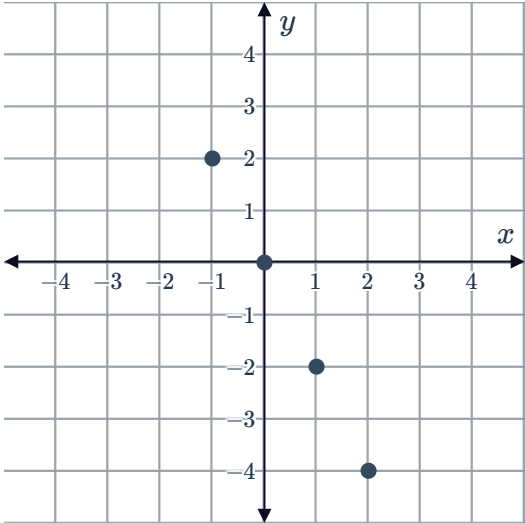


14

a

x	-1	0	1	2
$f(x)$	2	0	-2	-4

b

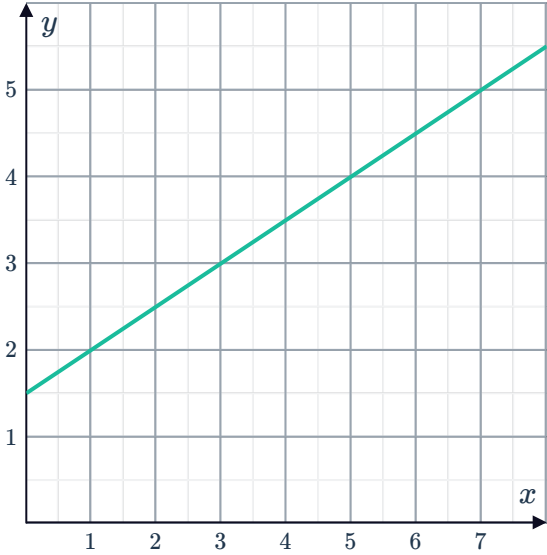


15

a

x	3	4	5	6	7
y	3	$\frac{7}{2}$	4	$\frac{9}{2}$	5

b



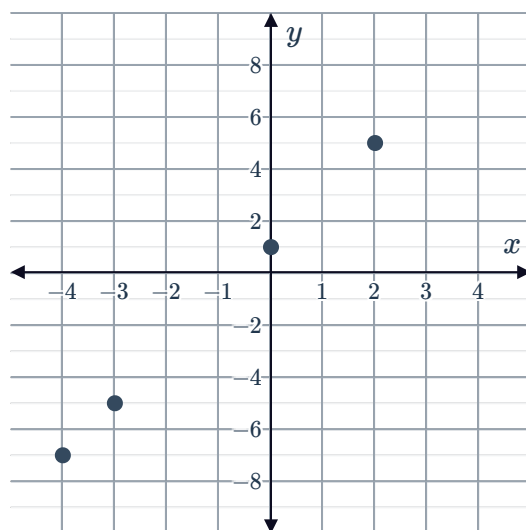
16 a $y = 2x + 1$



b

x	-4	-3	0	2
y	-7	-5	1	5

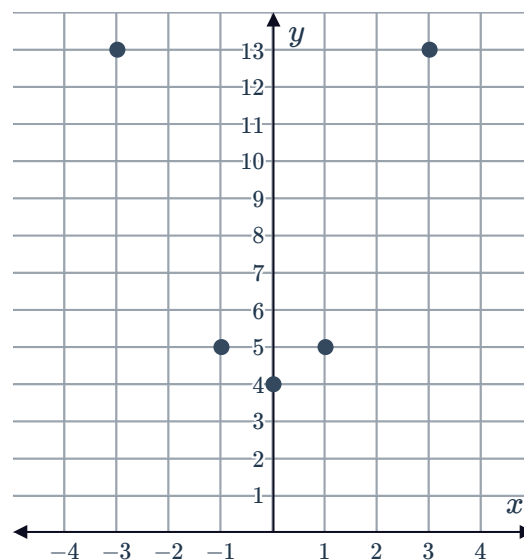
c



17 a

x	-3	-1	0	1	3
$f(x)$	13	5	4	5	13

b



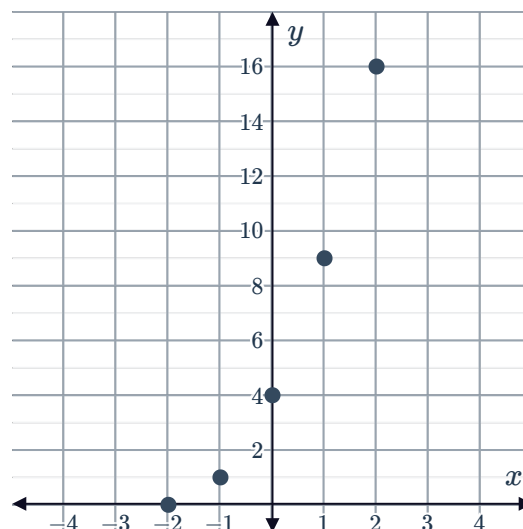
18

a

x	-2	-1	0	1	2
$f(x)$	0	1	4	9	16



b



19

a

i No

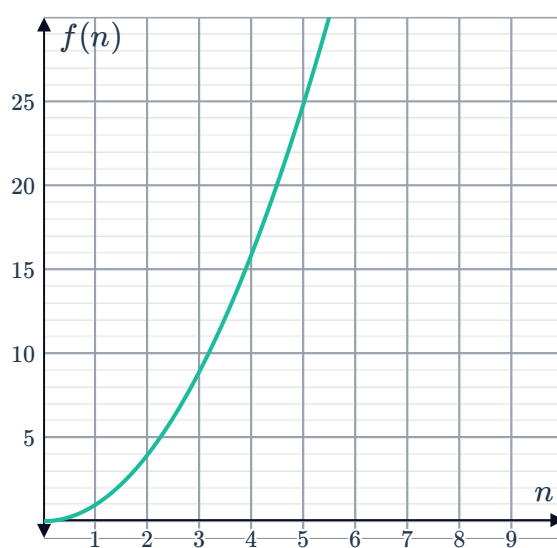
ii Yes

iii Yes

iv Yes

b $f(n) = n^2$

c



Rational functions

20 $(-\infty, -5) \cup (-5, \infty)$

21

a

i $(-\infty, 0) \cup (0, \infty)$ ii $(-\infty, 2) \cup (2, \infty)$

b

i $(-\infty, -6) \cup (-6, 6) \cup (6, \infty)$ ii $(-\infty, \infty)$



Applications

22

a $0 \leq t \leq 5$

c $0 \leq t \leq 3, 31 \leq t \leq 34$

b $25 \leq T \leq 80$

d $0 \leq t \leq 3$

23

a $0 \leq x \leq 7$

b 30

c No

24

a
$$f(x) = \begin{cases} -8, & 0 \leq x < 1 \\ -15, & 1 \leq x < 4 \\ -23, & 4 \leq x \leq 10 \end{cases}$$

c $-23^\circ C$

b $0 \leq x \leq 10$

d 10 am